

IMDb Web Automation Testing Framework

Project Name

IMDb Functional Testing Automation Framework Using Playwright

Created By

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1. Project Overview

This project is an end-to-end web automation testing framework developed using **Playwright with JavaScript** to validate the functionality, navigation, content visibility, and user experience of multiple services available on IMDb.

The framework focuses on testing critical user journeys across various IMDb sections and ensures that key features are working correctly through automated validation.

A total of **7 IMDb services** were automated and tested , covering homepage functionality, movie listings, celebrity information, events, recommendations, showtimes, and biography pages.

2. Objectives

- Automate functional testing of IMDb services.
 - Validate UI elements and navigation flows.
 - Verify content availability and correctness.
 - Ensure page stability and responsiveness.
 - Reduce manual testing effort.
 - Generate automated execution reports.
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3. Technologies Used

Technology	Purpose
JavaScript	Test scripting language
Playwright	Browser automation framework
Node.js	Runtime environment
Visual Studio Code	IDE
Git	Version control
GitHub	Source code repository
GitHub Actions	CI/CD Execution
Allure Report	Test reporting

4. Project Folder Structure

CAPSTONE PROJECT WIPRO

```
|
├── .github
│   └── workflows
│       └── playwright.yml
|
├── allure-results
├── playwright-report
├── test-results
├── node_modules
|
├── tests
│   ├── imdb_homepage.spec.js
│   ├── imdb_top250movies.spec.js
│   ├── imdb_popularCelebs.spec.js
│   ├── imdb_whatToWatch.spec.js
│   ├── imdb_events.spec.js
│   ├── imdb_SatyajitRay.spec.js
│   └── imdb_showtimes&tickets.spec.js
|
├── playwright.config.js
├── package.json
├── package-lock.json
├── README.md
└── Projects Documents
    ├── Project_report.pdf
    └── IMDb_testing_project.pdf
```

5. IMDb Services Covered

Service No.	IMDb Service
1	IMDb Homepage
2	IMDb Top 250 Movies
3	IMDb Popular Celebrities
4	IMDb What To Watch
5	IMDb Events
6	IMDb Satyajit Ray Profile
7	IMDb Showtimes & Tickets

6. Service-wise Test Cases

Service 1 – IMDb Homepage

Test Case	Business Value
Suggestion search input accepts text	High
IMDb homepage loads successfully	High
Open navigation drawer	Medium
Watchlist opens correctly	High
Most popular celebrities page loads	Medium
Language selection works	Medium

Test Case	Business Value
IMDb Pro page loads	Medium
Mobile apps page loads	Low
IMDb on social media	Low
Sign in is accessible	High

Service 2 – IMDb Top 250 Movies

Test Case	Business Value
Top 250 page opens	High
Movie list is visible	High
Ranking numbers are visible	High
Movie titles are displayed	High
Movie ratings are visible	High
Release year is displayed	Medium
Storyline is visible	Medium
Open movie details page	High
User reviews are visible	Medium
Ratings are valid	High
Search movie from Top 250	High
Page scroll works	Low
Page reload works properly	Low
No duplicate movie titles	High
Correct URL after opening movie	Medium

Service 3 – IMDb Popular Celebrities

Test Case	Business Value
IMDb popular celebs page launched	High
Anne Hathaway profile visible	Medium
Navigation bar opens	High
Sally Field biography contains correct information	Medium
Search bar works	High
Sydney Sweeney awards	Medium
Christopher Nolan known for	Medium
More to explore section	Low
Sally Field profile loads correctly	Medium
Grid view works	Medium
Emily Blunt profile picture	Low
Most popular celebs heading	High
Detailed view	Medium
Search category	Medium
Celeb news page loads	Medium

Service 4 – IMDb What To Watch

Test Case	Business Value
Page loads with main UI components	High
Navigation menu is functional	High
Search bar accepts input	High
Recommendation content visible	High
Page reload stability	Low
Mobile view works	Medium
No error messages	High
Page scroll works	Low
Links are present and clickable	High
Search functionality works	High
Fan Favorites section is visible	Medium
Category selector works	Medium
Heading is visible	Low

Service 5 – IMDb Events

Test Case	Business Value
Page loads with main UI components	High
12 Months Film Festival event loads correctly	Medium
Event links are functional	High
Search bar accepts input	High
Event details page loads	High
Mobile view works	Medium
Academy Awards winner info is correct	High
Mumbai Film Festival event page loads	Medium
Event details for Best Horror Baddy are visible	Medium
Cannes Film Festival event page loads	Medium

Service 6 – IMDb Satyajit Ray Profile

Test Case	Business Value
Heading is visible	High
Personal details are visible correctly	High
Known for section is working correctly	High
Awards and nominations contain correct information	High
Biography section is visible	High
Writer section is visible	Medium
Trivia section is visible	Medium
Quotes section contains correct information	Medium
Photos section loads correctly	Medium
Composer section is visible	Medium

Service 7 – IMDb Showtimes & Tickets

Test Case	Business Value
Movie page loads successfully	High
Showtimes section is available on page or redirect exists	High
Showtimes button is clickable	High
Navigation to showtimes page works	High
Showtimes page content loads	High
Location prompt appears or handled	High
Cinema listings exist	High
Showtimes are visible	High
Ticket links are visible	High
External ticket redirect works	High
Page has no broken UI	High
Back navigation works	Medium
Page title is correct	Medium
Multiple sections load on showtimes page	Medium
No critical elements missing	High

7. Test Coverage Summary

Metric	Count
Total Services Automated	7
Total Test Cases Executed	84
Framework	Playwright
Language	JavaScript
Browser Tested	Chromium
Reporting Tool	Allure Report
CI/CD Integration	GitHub Actions

8. Importance of the Project

Quality Assurance

Ensures critical IMDb features function correctly across multiple services.

Faster Regression Testing

Automated execution significantly reduces regression testing time compared to manual testing.

Improved Reliability

Detects broken links, navigation failures, missing content, and UI issues early.

Continuous Integration Support

GitHub Actions enables automated execution after every code push.

Better Reporting

Allure Reports provide detailed execution history, pass/fail statistics, and debugging information.

Scalability

The framework can be expanded easily to cover additional IMDb services and browsers.

Industry-Relevant Automation Skills

Demonstrates practical experience with:

- Playwright Automation
 - JavaScript Testing
 - UI Functional Testing
 - Test Reporting
 - CI/CD Pipelines
 - GitHub Integration
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9. CI/CD Pipeline (GitHub Actions)

This project uses **GitHub Actions** for Continuous Integration. Every push or pull request automatically triggers test execution to ensure code quality and stability.

Flow:

- Code pushed to GitHub
- GitHub Actions workflow triggers
- Node.js environment is set up
- Dependencies and Playwright browsers are installed
- Test cases are executed in headless mode
- Allure results are generated for reporting

Key Commands Used:

```
npm install
npx playwright install
npx playwright test
npm run allure:generate
npm run allure:open
```

Purpose:

- Automated regression testing
 - Faster feedback on code changes
 - Consistent test execution in CI environment
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10. Conclusion

The **IMDb Functional Testing Automation Framework** successfully automates testing for **7 major IMDb services** using **Playwright and JavaScript**. The framework validates core functionalities, navigation flows, content integrity, and user interactions through **84 automated test cases**, providing a scalable and maintainable solution for continuous web application testing.

